Olympic Medal Prediction Reproduction and Extension

Machine Learning Mini Project

Project aim

- Estimate a country's Summer Olympic medal count from earlier Olympic performance and broad national indicators.
- Reproduce the two-stage method in Brian C. Dobkowski's *2020 Summer Olympics Predictions Using Machine Learning*.
- Keep Rio 2016 as the final test set. This is necessary for a real reproduction because the paper also evaluates its method on 2016.
- Treat a 2020 forecast as an extension, not as part of the reproduction.

Data and preparation

- Olympic results come from Kaggle's 120 Years of Olympic History dataset. We use Summer Games from 1988 to 2016.
- World Bank World Development Indicators provide GDP, GDP per capita, population, population growth, land area, and each country's world share of GDP and population.
- We count one medal per country and event so a relay team is not counted several times.
- Additional features include athlete count, athlete share, total medals awarded, and medals won at the previous Games.
- IOC codes are matched to ISO3 codes. Rows with missing values are removed after the Olympic and World Bank data are joined.

Data split

- 1988-2008: model tuning with cross-validation.
- 2012: validation and model selection.
- 1988-2012: final training period before the 2016 test.

Method

- Stage 1 predicts whether a country will win any medal. A country classified as no-medal receives a prediction of zero.
- Stage 2 predicts the number of medals for countries that pass the first stage.
- The notebook compares logistic regression, SVM, Gaussian Naive Bayes, MLP, and random forest classifiers. It also tests linear, ridge, lasso, Poisson, SVM, random forest, and weighted regressors.
- Models are tuned with cross-validation. RMSE is the main measure, with a second loss that gives extra attention to the top ten medal-winning countries.

Results and observations

- The paper's preferred setup is Gaussian Naive Bayes followed by Ridge regression. We compare this with a plain linear-regression baseline.
- Our values may not match the paper exactly because World Bank indicators are revised, country-code matching differs, and not every tuning choice is reported.
- We test negative binomial regression because medal counts have many zeroes and a few very large values. The Poisson dispersion check is 1.56 and the estimated negative-binomial dispersion value is 0.215, supporting this extra check.
- An ablation without athlete-related features is included to check whether the result depends mainly on team size rather than economic indicators.

2020 extension

- Tokyo 2020 took place in 2021, so the actual medal table is now available.
- World Bank data for 2020 is also available. A forecast built only from information available before the Games can therefore be evaluated against the real results.
- This makes the future-work idea in the paper practical while keeping the 2016 reproduction fair and comparable.

Conclusion

The project first checks whether the published method can be reproduced on the same 2016 test set. It then explores more suitable count modelling and creates a clear path for a separately graded 2020 prediction.